

GRANT MCCONACHIE

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EDUCATION

Boston University, Boston, MA 2021 – Present

PhD, Biomedical Engineering | GPA: 3.93/4.00 | Advisor: Dr. Brian DePasquale

Oregon State University, Corvallis, OR 2016 – 2020

BS, Bioengineering | GPA: 3.85/4.00 | Summa Cum Laude

THESIS PROJECTS

Improve odorant-receptor interaction predictions using foundation models for underexplored *Aedes aegypti* receptors.

- Developed deep learning models to predict odor binding to mosquito olfactory receptors using high-dimensional protein-small molecule data and high-performance computing (HPC).
- Built multimodal low-rank adapted foundation models with PyTorch and HuggingFace frameworks.
- Contributions culminated in various posters and a manuscript currently under review at ICLR 2026.

GNN models for assessing the impact of individuality on collective dynamics in schooling fish.

- Developed custom graph neural network (GNN) models for latent graph inference to model collective behaviors of *Danionella translucida*.
- Implemented models using PyTorch and PyTorch Geometric frameworks.
- Work currently under review for presentation at COSYNE 2026.

Design novel GNN foundation models that exploit metabolic organization to generate olfaction specific odorant embeddings.

- Engineered multiple GNN foundation models incorporating pretext tasks tailored to capture olfactory-specific molecular features.
- Exploited metabolic pathway organization to inform embeddings, aiming to enhance molecular representations for downstream olfactory prediction tasks.

EXPERIENCE

Post Baccalaureate Fellow 2020 – 2021

National Cancer Institute | Frederick, MD

- Worked as an Automation Engineering Technician for the Frederick National Lab for Cancer Research.
- Developed software and automation scripts to enhance workflow efficiency in a high-throughput lab focused on discovering novel drug candidates from natural products.
- Developed a RESTful API, using flask, to control 6 automated chromatography machines, reducing manual workload by 90% and improving productivity.

Research Assistant 2019 – 2020

Oregon State University | Corvallis, OR

- Conducted research with Dr. Cory Simon to leverage mathematics, machine learning, statistical mechanics, and computer simulations to understand porous material.
- Aided in projects that used machine learning and data collection techniques to predict adsorption properties of metal organic frameworks (MOFs).
- Co-authored a paper published in the ACS Chemistry of Materials journal.

SKILLS

PyTorch, PyTorch geometric, HuggingFace, JAX, Jraph, Python, MATLAB, Cluster computing (SGE), API development, GNNs, Self-supervised learning, Transformers, Multimodal models, LoRA

POSTERS, PUBLICATIONS, AND TALKS

Learning the Language of Smell: Foundation Models for Protein-Odor Interactions.

Frontiers in NeuroAI Symposium 2025, Kempner Institute, USA. [Poster](#).

[Grant D. McConachie](#), Emily Duniec, Florence Guerina, Meg A. Younger, Brian DePasquale.

Improved Odor-Receptor Interaction Predictions via Self-Supervised Learning.

NAISys 2024, Cold Spring Harbor Laboratory, USA. [Poster](#).

[Grant D. McConachie](#), Emily Duniec, Meg A. Younger, Brian DePasquale.

A Graph Neural Network Self Supervised Learning Approach to Generate a Meaningful Chemical Latent Space for Olfactory Tasks.

ACChemS 2024, FL, USA. [Poster](#).

[Grant D. McConachie](#), Meg A. Younger, Brian DePasquale.

Graph neural network guided *in silico* deorphanization technique for olfactory receptors.

COSYNE 2024, Lisbon, Portugal. [Poster](#).

[Grant D. McConachie](#), Meg A. Younger, Brian DePasquale.

Screen for New Antimicrobial Natural Products from the NCI Program for Natural Product Discovery Prefractionated Extract Library.

ACS Infectious Diseases 2023. [Paper](#).

Lucero Martinez-Fructuoso, S. J. Ryan Arends, Vitor F. Freire, Jason R. Evans, Sean DeVries, Brian D. Peyser, Rhone K. Akee, Christopher C. Thornburg, Rohitesh Kumar, Susan Ensel, Gina M. Morgan, [Grant D. McConachie](#), Nathan Veeder, Leonard R. Duncan, Tanja Grkovic, Barry R. O'Keefe.

National Cancer Institute (NCI) Program for Natural Product Discovery: Exploring NCI-60 Screening Data of Natural Product Samples with Artificial Neural Networks.

ACS Omega 2023. [Paper](#).

Jason R Evans, Rhone K Akee, Shaurya Chanana, [Grant D. McConachie](#), Christopher C. Thornburg, Tanja Grkovic, Barry R. O'Keefe.

Recommendation System to Predict Missing Adsorption Properties of Nanoporous Materials.

Chemistry of Materials 2021. [Paper](#).

Arni Sturluson, Ali Raza, [Grant D. McConachie](#), Daniel W. Siderius, Xiaoli Z. Fern, Cory M. Simon.

HONORS AND AWARDS

National Science Foundation Graduate Research Fellowship (NSF GRF).

Awarded a 5-year fellowship to support my PhD research in applying state-of-the-art AI tools to gain an understanding of how the olfactory system encodes information.

Quantitative Biology & Physiology (QBP) Trainee.

Recipient of a prestigious 2-year training grant, supporting PhD candidates in computational biology and related fields.

SERVICE

Contributor for Optax

Added a normalized temperature scaled cross entropy loss ([NT-XENT](#)) to DeepMind's optimization library Optax.

Reviewer

Reviewer for ICML (2025), NeurIPS (2025), ICLR (2026), COSYNE (2026)

TEACHING EXPERIENCE

Control theory

Teaching Assistant | Boston University, Boston, MA | Fall 2026

Linear algebra

Teaching Assistant | Boston University, Boston, MA | Spring 2025

Graduate level intro to machine learning

Teaching Assistant | Boston University, Boston, MA | Spring 2024

Chemical process dynamics and simulation

Learning Assistant | Oregon State University, Corvallis, OR | Spring 2020